Extra Worksheet — Weeks 5-6

1. Six datapoints in $\mathbb{R}^3$ are

$$\mathbf{x}_1 = \begin{pmatrix}1\\3\\4\end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix}2\\2\\3\end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix}3\\4\\3\end{pmatrix}, \quad \mathbf{x}_4 = \begin{pmatrix}0\\0\\1\end{pmatrix}, \quad \mathbf{x}_5 = \begin{pmatrix}2\\2\\1\end{pmatrix}, \quad \mathbf{x}_6 = \begin{pmatrix}2\\3\\0\end{pmatrix}.$$

K-means begins with the three clusters

$$\mathcal{C}_1 = \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4\}, \qquad \mathcal{C}_2 = \{\mathbf{x}_5\}, \qquad \text{and} \qquad \mathcal{C}_3 = \{\mathbf{x}_6\}.$$

Run K-means until it terminates. For each round, state the three centroids and the clusters after reassignment. In any tied assignment, show the squared distances to the tied centroids and state how the tie-breaking rule is used.

2. Using the six datapoints from the previous question and Euclidean distance, run agglomerative hierarchical clustering separately with single linkage, complete linkage and average linkage.

(a) For each linkage rule, list every merge with its linkage distance and draw the resulting dendrogram.

(b) Compare the three dendrograms. Identify a cluster that is formed by one linkage rule but not by at least one of the other rules at the same stage.

(c) If one wants three clusters, what are the resulting partitions from the three dendrograms? Quantitatively assess the clustering quality of each distinct partition.

3. Let

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & 0 & 3 \\ 0 & 3 & 1 & -1 \\ 4 & 1 & -2 & 0 \\ 2 & 0 & 1 & 1 \\ -1 & 5 & 2 & 0 \\ 3 & -1 & 0 & 2 \end{pmatrix} \qquad \text{and} \qquad \mathbf{x} = \begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \end{pmatrix}.$$

State the shapes of $\mathbf{A}$, $\mathbf{A}^\mathsf{T}$ and $\mathbf{Ax}$. Compute $\mathbf{Ax}$ and $\mathbf{A}^\mathsf{T} \begin{pmatrix} 1 \\ 0 \\ -1 \\ 2 \\ 0 \\ 1 \end{pmatrix}$.

4. Define $T : \mathbb{R}^3 \to \mathbb{R}^4$ by

$$T\left(\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}\right) = \begin{pmatrix} 4x_3 - 3x_2 \\ 0 \\ x_1 - x_3 \\ (x_1 + 3x_2 + x_3)/5 \end{pmatrix}.$$

Write a matrix $\mathbf{B}$ such that $T(\mathbf{x}) = \mathbf{Bx}$ for every $\mathbf{x} \in \mathbb{R}^3$.

5. Is the map $S : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$S\left(\begin{pmatrix}x_1\\x_2\end{pmatrix}\right) = \begin{pmatrix} 5x_1 - 11x_2 + 1 \\ 3x_2 \end{pmatrix}$$

linear? Explain your answer.

6. Let

$$\mathbf{A} = \begin{pmatrix} 6 & -8 & 3 \\ 0 & -4 & 9 \end{pmatrix} \qquad \text{and} \qquad \mathbf{B} = \begin{pmatrix} 6 & 0 \\ 1 & 2 \\ -3 & -7 \end{pmatrix}.$$

(a) State the shapes of $\mathbf{AB}$ and $\mathbf{BA}$.

(b) Compute $\mathbf{AB}$ and $\mathbf{BA}$.

(c) If $\mathbf{x} \in \mathbb{R}^2$, describe in words the order in which the transformations are applied in $(\mathbf{AB})\mathbf{x}$.

7. The rows of the data matrix $\mathbf{X}$ are the transposes of four datapoints in $\mathbb{R}^2$:

$$\mathbf{X} = \begin{pmatrix} 1 & 2 \\ 0 & -1 \\ 3 & 1 \\ -4 & 5 \end{pmatrix}.$$

The matrix

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ -3 & 0 \end{pmatrix}$$

represents a linear map from $\mathbb{R}^2$ to $\mathbb{R}^2$. Compute the data matrix $\mathbf{Y}$ obtained by applying the above linear map to the datapoints in $\mathbf{X}$.

8. (a) Construct matrices $\mathbf{A} \in \mathbb{R}^{3\times 2}$ and $\mathbf{B} \in \mathbb{R}^{2\times 3}$ such that

$$\mathbf{BA} = \mathbf{I}_2 \qquad \text{but} \qquad \mathbf{AB} \neq \mathbf{I}_3.$$

Verify both claims for your matrices. (Reminder: $\mathbf{I}_d$ is the $d \times d$ identity matrix.)

(b) (*) Prove that it is impossible to construct matrices $\mathbf{C} \in \mathbb{R}^{3\times 2}$ and $\mathbf{D} \in \mathbb{R}^{2\times 3}$ such that

$$\mathbf{DC} = \mathbf{I}_2 \qquad \text{and} \qquad \mathbf{CD} = \mathbf{I}_3.$$

9. Write the following linear system as an augmented matrix and use Gaussian elimination to solve it.

$$\begin{aligned} x_1 + 2x_2 + x_3 + x_4 + 3x_5 &= 3, \\ 2x_1 + 4x_2 + 4x_4 - x_6 &= 3, \\ x_3 - x_4 + 3x_5 + 3x_6 &= -1. \end{aligned}$$

10. (*) Suppose a linear system has m equations and d variables, where $m < d$. Show that it is either inconsistent or has infinitely many solutions.

11. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 3 & 7 & 3 & 8 \end{pmatrix} \qquad \text{and} \qquad \mathbf{B} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 3 & 7 & 3 & 7 \end{pmatrix}.$$

For each of the above matrices, determine whether it has an inverse. If yes, compute the inverse.

12. (*) Suppose $\mathbf{A} \in \mathbb{R}^{d\times d}$, and $\mathbf{I}_d$ is the $d \times d$ identity matrix. If $\mathbf{A}^2$ is the zero matrix (i.e., every entry in $\mathbf{A}^2$ is zero), show that $\mathbf{I}_d - \mathbf{A}$ has an inverse. (Reminder: $\mathbf{A}^2$ is same as the matrix product $\mathbf{AA}$.)

13. Let

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 2 \\ 1 \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ -2 \\ 1 \end{pmatrix}, \qquad \mathbf{v}_3 = \begin{pmatrix} 0 \\ 2 \\ -4 \\ 3 \end{pmatrix}, \qquad \mathbf{q} = \begin{pmatrix} 2 \\ -2 \\ 8 \\ 5 \end{pmatrix}.$$

Determine whether $\mathbf{q} \in \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. If it is, find coefficients that express $\mathbf{q}$ as a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$.

14. For each set below, determine whether it is a subspace of $\mathbb{R}^3$. Justify your answer using the subspace conditions.

$$\mathcal{S} = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \;\middle|\; x - 2y + z = 0 \right\}, \qquad \mathcal{T} = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \;\middle|\; x - 2y + z = 1 \right\}.$$

15. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 2 & 1 \end{pmatrix}.$$

State the ambient spaces that contain $\text{Col}(\mathbf{A})$ and $\text{Nul}(\mathbf{A})$. Show that $\text{Col}(\mathbf{A})$ is the same as its ambient space, whereas $\text{Nul}(\mathbf{A})$ is a proper subset of its ambient space.